Page 2: The Nature of Light and Its Travel

What is Light?

Light is a form of energy that enables us to see the world around us. It is an electromagnetic wave that does not require a medium to travel (it can travel through a vacuum).

Key Properties of Light:

- It travels in a straight line in a homogeneous transparent medium. This is known as **rectilinear propagation of light**.
- Its speed in a vacuum is the ultimate speed limit of the universe: **3×10^8 m/s** (approximately 300,000 km/s).
- It carries energy.

Important Definitions for Your Exam:

- **Luminous Objects:** Objects that emit their own light.
    - *Examples:* Sun, stars, candle flame, electric bulb.
- **Non-Luminous Objects:** Objects that do not emit their own light. They become visible only when light from a luminous object falls on them and is reflected into our eyes.
    - *Examples:* Moon, planets, table, chair, book.
- **Ray of Light:** The straight line path along which light travels. Represented by a straight line with an arrow showing the direction.
- **Beam of Light:** A bundle of light rays.
    - **Parallel Beam:** Rays are parallel to each other (e.g., light from the sun).
    - **Convergent Beam:** Rays meet at a point (e.g., light focused by a convex lens).
    - **Divergent Beam:** Rays spread out from a point (e.g., light from a candle).

Page 3: What Happens When Light Encounters an Object?

When light traveling in one medium (like air) falls on another medium (like glass or water), one or more of the following three phenomena occur:

1. Reflection
The light bounces back into the same medium. This is what allows us to see non-luminous objects. You look into a mirror, and the light bounces back to your eyes.

2. Refraction
The light passes from one medium into the other, but its path bends at the boundary. This happens because the speed of light changes as it enters the new medium.

- *Example:* A pencil appearing bent in a glass of water.

3. Absorption
The light is absorbed by the medium and its energy is converted into heat. This is why dark objects get hot in the sun.

For Class 10, we focus on the first two phenomena: **Reflection** and **Refraction**.

Page 4: Reflection of Light – The Basics

Definition: The phenomenon of bouncing back of light from the surface of an object is called reflection.

Laws of Reflection (The Most Important Rule!)
These two laws hold true for all types of reflection (whether the surface is plane or curved):

1. **The First Law:** The angle of incidence is always equal to the angle of reflection.
    - **$\angle i = \angle r$**
2. **The Second Law:** The incident ray, the reflected ray, and the normal (the perpendicular line) to the surface at the point of incidence, all lie in the same plane.

Key Terms to Know:

- **Incident Ray:** The incoming ray that strikes the surface.
- **Point of Incidence:** The point on the surface where the ray strikes.
- **Reflected Ray:** The ray that bounces off the surface.
- **Normal:** An imaginary line perpendicular to the surface at the point of incidence.
- **Angle of Incidence (i):** The angle between the incident ray and the normal.
- **Angle of Reflection (r):** The angle between the reflected ray and the normal.

Page 5: Types of Reflection

Based on the nature of the reflecting surface, reflection is of two types:

1. Regular (or Specular) Reflection

- **Surface:** Occurs on a smooth, polished surface (like a plane mirror or still water).

- **Behavior:** Parallel incident rays remain parallel after reflection.
- **Result:** A clear, sharp image is formed.
- *Example:* Your image in a mirror.

2. Irregular (or Diffused) Reflection

- **Surface:** Occurs on a rough, uneven surface (like a wall, paper, or cloth).
- **Behavior:** Parallel incident rays are reflected in many different directions.
- **Result:** No clear image is formed, but the surface becomes visible.
- *Example:* You can see a chair in a room from any direction because of diffused reflection.

Important Concept:
Diffused reflection does **not** violate the laws of reflection. Each individual ray obeys the laws ($\angle i = \angle r$), but because the surface is uneven, the normals at each point are in different directions. Therefore, the reflected rays go in different directions.

Page 6: Spherical Mirrors – Terminology

A spherical mirror is a mirror whose reflecting surface is a part of a hollow sphere.

Types of Spherical Mirrors:

1. **Concave Mirror (Converging Mirror):** Reflection takes place on the *inner* curved surface (the bulging-in part). It converges light rays.
2. **Convex Mirror (Diverging Mirror):** Reflection takes place on the *outer* curved surface (the bulging-out part). It diverges light rays.

Key Terms (Memorize these for exams):

- **Center of Curvature (C):** The center of the sphere of which the mirror is a part.
- **Radius of Curvature (R):** The radius of the sphere. (R = distance from pole to center of curvature).
- **Pole (P):** The center point of the mirror's surface.
- **Principal Axis:** The imaginary line passing through the pole and the center of curvature.
- **Aperture:** The diameter of the mirror's reflecting surface. It determines how much light the mirror can gather.

- **Focus (F):** For a concave mirror, it's the point on the principal axis where parallel light rays converge after reflection. For a convex mirror, it's the point from which parallel rays appear to diverge after reflection.
- **Focal Length (f):** The distance between the pole and the focus. For spherical mirrors, **f = R/2**.

Page 7: Image Formation by Spherical Mirrors

Understanding the Terms:

- **Real Image:** Formed when light rays actually meet. It can be obtained on a screen. It is always inverted.
- **Virtual Image:** Formed when light rays appear to meet (but don't actually meet). It cannot be obtained on a screen. It is always erect.

Conceptual Summary for Concave Mirror:
The position of the object determines the nature of the image.

- **Object at infinity:** Image is real, inverted, and highly diminished, formed at F.
- **Object beyond C:** Image is real, inverted, and diminished, formed between C and F.
- **Object at C:** Image is real, inverted, and same size, formed at C.
- **Object between C and F:** Image is real, inverted, and enlarged, formed beyond C.
- **Object at F:** Image is real, inverted, and highly enlarged, formed at infinity.
- **Object between P and F:** Image is virtual, erect, and enlarged, formed behind the mirror. (This is the principle behind a shaving/makeup mirror).

Conceptual Summary for Convex Mirror:

- **For all positions of the object (in front of the mirror):** The image is always **virtual, erect, and diminished**. It is formed behind the mirror between P and F. This wide field of view makes it ideal for rear-view mirrors in vehicles.

Uses:

- **Concave:** Torches, headlights, shaving mirrors, solar furnaces.
- **Convex:** Rear-view mirrors in vehicles.

Page 8: Refraction of Light – The Basics

Definition: The phenomenon of change in the direction of light when it passes from one transparent medium to another is called refraction.

What causes refraction?
The bending of light occurs because **the speed of light is different in different media**.

- Light travels fastest in a vacuum.
- It travels slower in any other medium (like air, water, or glass).

The Rules of Refraction (What happens to the light ray?)

- **When light travels from a rarer to a denser medium:**
  - Speed decreases.
  - The ray bends **towards the normal**.
  - *Example:* Air (rarer) to Glass (denser).
- **When light travels from a denser to a rarer medium:**
  - Speed increases.
  - The ray bends **away from the normal**.
  - *Example:* Glass (denser) to Air (rarer).

Optical Density vs. Mass Density:
Do not confuse optical density with physical density.

- A medium which has a higher refractive index (slower speed of light) is called **optically denser**.
- *Example:* Kerosene, though it floats on water (physically less dense), is optically denser than water because it slows light down more.

Page 9: The Laws of Refraction

1. The First Law (The Planar Law)
The incident ray, the refracted ray, and the normal at the point of incidence, all lie in the same plane.

2. The Second Law (Snell's Law of Refraction)
This is the quantitative law. It states that for a given pair of media, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant.

Snell's Law Formula:
$n = \sin i / \sin r$

Where 'n' is the **refractive index** of the second medium with respect to the first.

What is Refractive Index?
It is a measure of how much a medium can bend light. It is the ratio of the speed of light in a vacuum (or air) to the speed of light in that medium.

- **Absolute Refractive Index (n):** n = Speed of light in vacuum / Speed of light in medium = c/v
- Since the speed of light in a medium is always less than that in a vacuum, **n is always greater than 1** for any medium other than vacuum.
- **Relative Refractive Index:** When we talk about the refractive index of medium 2 with respect to medium 1.
    - n_{21} = Speed of light in medium 1 / Speed of light in medium 2 = v_1/v_2

Page 10: Lenses and At-Home Phenomena

Lenses
A lens is a transparent medium bound by two spherical surfaces.

- **Convex Lens (Converging Lens):** Thicker in the middle, thinner at the edges. It converges parallel light rays to a point (focus).
- **Concave Lens (Diverging Lens):** Thinner in the middle, thicker at the edges. It diverges parallel light rays, which appear to come from a point (focus).

Summary of Image Formation:

- **Convex Lens:** Can form both real and virtual images depending on the object's position. (Used in magnifying glass, cameras, human eye).
- **Concave Lens:** Always forms a **virtual, erect, and diminished** image for all positions of the object. (Used in spectacles to correct myopia).

Phenomena in Nature due to Refraction:

1. **A pencil appears bent in water:** Light from the submerged part of the pencil comes from water (denser) to air (rarer) and bends away from the normal, making the pencil appear raised.
2. **The twinkling of stars:** Due to atmospheric refraction. The apparent position of the star keeps changing slightly because of the constant movement of air layers with different refractive indices.

3. **Sun seen before actual sunrise:** When the sun is just below the horizon, its light rays bend due to atmospheric refraction and reach our eyes, making the sun appear to have risen a few minutes earlier.